SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Problems

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Problems
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Perugu Siva Kumar	**Reg. No.:**	192311088
Section / Batch:	CSE	**Date:**	30/10/26

Code of Conduct

I _____Perugu Siva Kumar(192311088)__ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __P.SIVA KUMAR____

Instructions

- This test contains 20 questions.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio ($10 \cdot \log 10$ of the ratio).

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Q1. Duplex voice channels from a 45 MHz spectrum auction

A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Given

- Total spectrum, B_total = 45 MHz = 45,000 kHz
- Bandwidth of one simplex channel, B_c = 25 kHz

Concept / Formula used

```
Simplex channels = B_total / B_c
1 full-duplex channel = 2 simplex channels (one forward + one reverse link)
Duplex channels = Simplex channels / 2
```

Step-by-step solution

Step 1. Number of simplex channels = 45,000 kHz / 25 kHz = 1,800.

Step 2. A full-duplex voice call needs a forward (downlink) and a reverse (uplink) channel, so each duplex channel uses 2 × 25 kHz = 50 kHz.

Step 3. Duplex channels = 1,800 / 2 = 900 (or 45,000 kHz / 50 kHz = 900).

Answer: 900 full-duplex voice channels (1,800 simplex channels)

Additional information

- In FDD systems the forward and reverse channels are separated by a fixed duplex spacing; for example, the AMPS band is 25 MHz forward + 25 MHz reverse.
- If the 45 MHz were split into two equal halves for uplink and downlink (22.5 MHz each), the result is the same: 22,500 / 25 = 900 channels.
- This is a raw upper bound. Guard bands and control channels reduce the number of channels actually available for traffic.

Q2. Channels allocated to one cell in a 7-cell cluster

A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Given

- Total channels, S = 560
- Cluster size, N = 7 cells

Concept / Formula used

```
Channels per cell, k = S / N
Total channels S = k × N
```

Step-by-step solution

Step 1. All 560 channels are divided among the 7 cells of a cluster; no channel is reused inside the cluster.

Step 2. k = 560 / 7 = 80 channels per cell.

Answer: 80 channels per cell

Additional information

- Each cluster uses all 560 channels; the same set is reused in the next cluster, so a network with M clusters has a total capacity C = M × k × N = M × S.
- For a given S, a smaller N gives more channels per cell (higher capacity) but a shorter reuse distance D = R√(3N) and therefore more co-channel interference.
- N = 7 is the classic choice for GSM/AMPS, because it gives a co-channel reuse ratio D/R = √21 ≈ 4.58.

Q3. Simultaneous voice channels in a GSM deployment

A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Given

- Number of carriers = 100 (spacing 200 kHz)
- Time slots per carrier = 8

Concept / Formula used

```
Total channels = number of carriers × slots per carrier (FDMA + TDMA)
```

Step-by-step solution

Step 1. Each carrier is divided in time into 8 slots; each slot carries one full-rate voice call.

Step 2. Total channels = 100 × 8 = 800.

Step 3. Check on bandwidth used: 100 × 200 kHz = 20 MHz, so the deployment needs 20 MHz of spectrum (per direction).

Answer: 800 simultaneous voice channels

Additional information

- GSM combines FDMA (200 kHz carriers) with TDMA (8 slots). That is why the same spectrum carries 8 times more calls than a pure FDMA system.
- In practice one or two slots of the BCCH carrier are used for signalling, and some slots are used for SDCCH, so the real number of traffic channels (TCH) is smaller.
- With half-rate speech coding each slot is shared by two users, theoretically giving 1,600 half-rate channels.

Q4. Power ratio between base station and handset in dB

During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

Given

- Base station power, P_BS = 80 W
- Handset power, P_MS = 4 W

Concept / Formula used

```
Ratio (dB) = 10 · log10(P1 / P2)
```

Step-by-step solution

Step 1. Ratio = 80 / 4 = 20.

Step 2. Ratio (dB) = 10 · log10(20) = 10 × 1.3010.

Step 3. Ratio (dB) = 13.01 dB.

Answer: ≈ 13 dB (the base station transmits 20 times more power)

Additional information

- Useful dB anchors: 3 dB ≈ ×2, 10 dB = ×10, 13 dB ≈ ×20 (10 dB + 3 dB), so 13 dB is exactly what you expect from 10 × 2.
- Handset power is limited by battery life and SAR safety limits; base stations have mains power and larger antennas.
- Power control keeps the received power at the base station roughly equal for near and far users, which is critical in CDMA systems (near-far problem).

Q5. Carrier-to-Interference ratio from a drive test

A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Given

- Carrier (serving signal) C = −60 dBm
- Interference I = −92 dBm

Concept / Formula used

```
C/I (dB) = C (dBm) − I (dBm)   (subtracting dBm values gives dB)
```

Step-by-step solution

Step 1. C/I = (−60) − (−92) = −60 + 92.

Step 2. C/I = 32 dB.

Step 3. As a linear ratio: 10^(32/10) = 10^3.2 ≈ 1,585 (the carrier is about 1,585 times stronger than the interference).

Answer: C/I = 32 dB

Additional information

- GSM needs a C/I of about 9 dB (typically 12 dB with margin); AMPS about 18 dB. At 32 dB the link quality is therefore excellent.
- Required C/I is directly related to cluster size: for a hexagonal layout with six co-channel interferers, C/I ≈ (D/R)^n / 6 = (3N)^(n/2)/6.
- C/I ignores thermal noise. The related SINR includes noise: SINR = C / (I + N).

Q6. Channels per cell from 42 MHz with 300 kHz channels and N = 7

An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Given

- Total spectrum = 42 MHz = 42,000 kHz
- Channel bandwidth = 300 kHz
- Cluster size N = 7

Concept / Formula used

```
Total channels S = B_total / B_c
Channels per cell k = S / N
```

Step-by-step solution

Step 1. S = 42,000 kHz / 300 kHz = 140 channels.

Step 2. k = S / N = 140 / 7 = 20 channels per cell.

Answer: 20 channels per cell (140 channels in total)

Additional information

- Here each 'channel' is taken as one channel of 300 kHz (simplex). If the system were FDD with paired channels, you would divide by 2 first to get 70 duplex channels and then about 10 per cell.
- Increasing the reuse (smaller N) would raise k: for N = 4 each cell would get 35 channels.
- Capacity of the whole system scales with the number of clusters, because each cluster re-uses the same 140 channels.

Q7. Received power when distance is doubled (path-loss exponent 4)

A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Given

- Received power at original distance, P_1 = 64 W (as stated: power delivered to the receiver)
- New distance $d_2 = 2\ d_1$
- Path-loss exponent n = 4

Concept / Formula used

$P_r \propto 1 / d^n$

$P_2 = P_1 \times (d_1 / d_2)^n$

Step-by-step solution

Step 1. $P_2 = 64 \times (d_1 / 2d_1)^4 = 64 \times (1/2)^4$.

Step 2. $(1/2)^4 = 1/16$.

Step 3. $P_2 = 64 / 16 = 4$ W.

Step 4. Loss in dB = 10 log10(16) = 12 dB (each doubling of distance costs 3n dB = 12 dB).

Answer: 4 W (a drop of 12 dB)

Additional information

- Interpretation used: the 64 W is the power received at the original distance. This is the only reading that gives a meaningful answer without antenna gains, wavelength or reference-distance data.
- Typical path-loss exponents: free space 2; urban cellular 2.7–3.5; shadowed urban 3–5; in-building line of sight 1.6–1.8; obstructed in-building 4–6.
- The steep n = 4 law is the reason mobile cells can be small and frequencies can be reused after a short distance.

Q8. Transmit-power scaling when the cell radius is doubled

Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Given

- Path-loss exponent n = 2 (inverse-square law)
- New radius $R_2 = 2 R_1$
- Required received power (SNR) unchanged

Concept / Formula used

$P_r = P_t \cdot K / R^n$

For the same P_r: $P_{t2} / P_{t1} = (R_2 / R_1)^n$

Step-by-step solution

Step 1. $P_{t2} / P_{t1} = (2R_1 / R_1)^2$.

Step 2. $= 2^2 = 4$.

Step 3. In dB: 10 log10(4) = 6 dB.

Answer: Transmit power must be increased by a factor of 4 (+6 dB)

Additional information

- For a more realistic urban exponent n = 4, doubling the radius would need $2^4 = 16$ times (12 dB) more power. That is why operators add sites instead of raising power.
- Higher transmit power also increases interference to co-channel cells, so coverage extension by power alone is limited.
- Coverage area grows with R^2, so doubling R covers 4 times the area – a factor of 4 in power for a factor of 4 in area under inverse-square law.

Q9. Total channels in a cluster of 4 cells

A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Given

- Cluster size N = 4
- Channels per cell k = 15

Concept / Formula used

$S = k \times N$

Step-by-step solution

Step 1. S = 15 × 4 = 60 channels.

Answer: 60 channels in the cluster

Additional information

- The 60 channels are all different within the cluster; the next cluster reuses exactly these 60.
- Total system capacity with M clusters: C = M × 60. For example, 10 clusters offer 600 simultaneous channels.
- Cluster size must be $i^2 + i \cdot j + j^2$ (1, 3, 4, 7, 9, 12, …). N = 4 is valid with i = 2, j = 0.

Q10. Doppler shift at 100 km/h on 900 MHz

A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Given

- Speed v = 100 km/h
- Carrier frequency f_c = 900 MHz
- Speed of light c = 3×10^8 m/s

Concept / Formula used

$f_d = (v / \lambda) \cos\theta$, maximum when $\theta = 0$

$\lambda = c / f_c$

$f_{d,max} = v \cdot f_c / c$

Step-by-step solution

Step 1. Convert speed: v = 100 × 1000 / 3600 = 27.78 m/s.

Step 2. Wavelength: $\lambda = 3 \times 10^8 / 900 \times 10^6$ = 0.3333 m.

Step 3. f_d = 27.78 / 0.3333 = 83.3 Hz.

Answer: Doppler shift ≈ 83 Hz (maximum, when the vehicle moves directly towards or away from the base station)

Additional information

- The received frequency becomes 900 MHz ± 83 Hz depending on whether the vehicle approaches (+) or recedes (−).

- Doppler spread causes rapid fading; coherence time T_c ≈ 0.423 / f_d ≈ 5 ms here, so the channel changes within a few ms.
- GSM equalisers and OFDM subcarrier spacing are designed to tolerate this level; at high-speed rail (350 km/h) and higher carriers the shift grows to several hundred Hz.

Q11. Slot duration in a 5 ms TDMA frame with 8 slots

A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Given

- Frame duration T_f = 5 ms
- Slots per frame = 8

Concept / Formula used

```
Slot duration T_s = T_f / number of slots
```

Step-by-step solution

Step 1. T_s = 5 ms / 8 = 0.625 ms.

Step 2. In microseconds: 0.625 ms = 625 µs.

Answer: 0.625 ms (625 µs) per slot

Additional information

- Real GSM uses a 4.615 ms frame (120/26 ms) with 8 slots of 0.577 ms (577 µs) – see Q35 for the 4.8 ms variant.
- Each user transmits only once per frame, so a user's duty cycle is 1/8 = 12.5%, which saves battery power.
- Guard time inside each slot (about 8.25 bit periods in GSM) prevents overlap of bursts from users at different distances.

Q12. Number of GSM carriers in 24 MHz

Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Given

- Allocated bandwidth = 24 MHz = 24,000 kHz
- Carrier spacing = 200 kHz

Concept / Formula used

```
Number of carriers = Total bandwidth / carrier spacing
```

Step-by-step solution

Step 1. Carriers = 24,000 kHz / 200 kHz = 120.

Answer: 120 carrier frequencies

Additional information

- The question asks for carriers, so no halving for duplex is needed. If 24 MHz is the total for both uplink and downlink, there would be 60 duplex carriers (12 MHz per direction ÷ 200 kHz).
- Each carrier carries 8 TDMA slots, so 120 carriers could carry up to 120 × 8 = 960 physical channels.

- In real GSM-900, 25 MHz per direction yields 124 usable carriers because one 200 kHz carrier is used as guard band.

Q13. Received power at base station after 33 dB path loss

A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Given

- Transmit power P_t = 8 W
- Path loss L = 33 dB

Concept / Formula used

```
P(dBm) = 10 log10(P in mW)
P_r (dBm) = P_t (dBm) − L (dB)
```

Step-by-step solution

Step 1. Convert to mW: 8 W = 8,000 mW.

Step 2. P_t (dBm) = 10 log10(8,000) = 10 × 3.9031 = 39.03 dBm (equivalently 30 dBm + 10 log10 8 = 30 + 9.03).

Step 3. P_r = 39.03 − 33 = 6.03 dBm.

Step 4. In linear units: 10^0.603 ≈ 4.0 mW.

Answer: ≈ 6 dBm (≈ 4 mW)

Additional information

- Gains of the antennas and cable losses were not specified and are taken as 0 dB. In a full link budget: P_r = P_t + G_t + G_r − L − losses.
- Received power of +6 dBm is unrealistically strong for a mobile network (receivers usually see −50 to −110 dBm); the value reflects the very small path loss given in the problem.
- Remember 30 dBm = 1 W and 0 dBm = 1 mW.

Q14. Free-space path loss at 1900 MHz over 2.5 km

A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Given

- Frequency f = 1900 MHz
- Distance d = 2.5 km = 2,500 m
- Reference distance d_0 = 1 m
- c = 3 × 10^8 m/s

Concept / Formula used

```
λ = c / f
PL(d₀) = 20 log10(4π d₀ / λ)
PL(d) = PL(d₀) + 20 log10(d / d₀)   (free space, n = 2)
Shortcut: FSPL (dB) = 32.44 + 20 log10 d(km) + 20 log10 f(MHz)
```

Step-by-step solution

Step 1. λ = 3×10^8 / 1.9×10^9 = 0.1579 m.

Step 2. Loss at the 1 m reference: $PL(d_0) = 20 \log10(4\pi \times 1 / 0.1579) = 20 \log10(79.59) = 38.02$ dB.

Step 3. Extra loss from 1 m to 2,500 m: $20 \log10(2{,}500 / 1) = 20 \times 3.3979 = 67.96$ dB.

Step 4. PL(2.5 km) = 38.02 + 67.96 = 105.98 dB.

Step 5. Check with shortcut: 32.44 + 20 log10(2.5) + 20 log10(1900) = 32.44 + 7.96 + 65.58 = 105.98 dB.

Answer: Free-space path loss ≈ 106 dB

Additional information

- Free-space loss increases by 6 dB when distance or frequency is doubled.
- Real urban macro-cell loss is higher than free-space because of buildings and ground reflection (n = 3 to 4), so the real value over 2.5 km could be 130–150 dB.
- The 1 m reference is used in the log-distance model: $PL(d) = PL(d_0) + 10\, n \log10(d/d_0) + X_\sigma$.

Q15. Total call-handling capacity of a 6-cell cluster

A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Given

- Cells in cluster = 6
- Capacity per cell = 250 simultaneous calls

Concept / Formula used

```
Cluster capacity = number of cells × capacity per cell
```

Step-by-step solution

Step 1. Capacity = 6 × 250 = 1,500 simultaneous calls.

Answer: 1,500 simultaneous calls

Additional information

- This is a peak value: it assumes every cell is fully loaded at the same time, which rarely occurs because traffic varies in space and time.
- Real planning keeps the blocking probability low (for example 2%) using the Erlang-B formula, so the offered traffic is lower than the channel count.
- Capacity can be increased by cell splitting, sectoring and additional carriers.

Q16. Active calls in a switch (Little's law)

A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Given

- Call arrival rate λ = 4,000 calls/s
- Average holding time h = 3 min = 180 s

Concept / Formula used

```
Little's law: N = λ × h  (average number in the system = arrival rate × average time in the system)
Equivalent to traffic A (Erlangs) = λ × h
```

Step-by-step solution

Step 1. Convert holding time: 3 min × 60 = 180 s.

Step 2. N = 4,000 × 180 = 720,000 calls.

Answer: 720,000 simultaneous (active) calls (= 720,000 Erlangs)

Additional information

- Since each active call occupies one circuit/bearer, the switch must provide at least 720,000 connections to avoid blocking at this load (more if the blocking target is strict).
- Little's law is valid for any arrival pattern and any holding-time distribution, as long as the system is stable.
- Halving the holding time to 90 s halves the active calls to 360,000.

Q17. Total bandwidth for 150 channels of 25 kHz

Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Given

- Channels = 150
- Bandwidth per channel = 25 kHz

Concept / Formula used

```
Total bandwidth = channels × bandwidth per channel
```

Step-by-step solution

Step 1. B = 150 × 25 kHz = 3,750 kHz.

Step 2. 3,750 kHz = 3.75 MHz.

Answer: 3.75 MHz (3,750 kHz)

Additional information

- If the channels are duplex pairs, the total spectrum consumed is 2 × 3.75 = 7.5 MHz (3.75 MHz uplink + 3.75 MHz downlink).
- Guard bands between the channels would add to this figure.
- Narrow 25 kHz channels are typical of PMR/analogue systems; GSM uses 200 kHz and LTE 1.4–20 MHz.

Q18. Total bandwidth lost to guard bands

Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Given

- Guard band size = 40 kHz
- Number of bands = 30

Concept / Formula used

```
Wasted bandwidth = number of guard bands × guard-band width
```

Step-by-step solution

Step 1. Taking one 40 kHz guard band for each of the 30 bands (as stated in the question): 30 × 40 kHz = 1,200 kHz.

Step 2. 1,200 kHz = 1.2 MHz.

Answer: 1.2 MHz wasted (1,200 kHz)

Additional information

- Strict geometry: 30 bands in a row have only 29 gaps between them, giving 29 × 40 = 1,160 kHz (1.16 MHz). If a guard band is also placed at both ends the count is 31 × 40 = 1,240 kHz. The intended textbook answer is 1.2 MHz.
- Guard bands reduce spectral efficiency; filter quality, OFDM windowing and careful frequency planning allow narrower guards.
- LTE uses about 10% of the channel bandwidth as guard (e.g., 18 MHz occupied in a 20 MHz channel).

Q19. Handoffs for a vehicle at 90 km/h in 1.5 km-radius cells

A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Given

- Speed v = 90 km/h
- Time = 1 hour
- Cell radius R = 1.5 km

Concept / Formula used

```
Distance travelled = v × t
Crossing distance per cell = 2R
Handoffs ≈ distance / crossing distance
```

Step-by-step solution

Step 1. Distance in 1 h = 90 km/h × 1 h = 90 km.

Step 2. Cell diameter = 2 × 1.5 = 3 km.

Step 3. Handoffs = 90 / 3 = 30.

Answer: ≈ 30 handoffs per hour (one every 2 minutes)

Additional information

- Time spent per cell = 3 km / 90 km/h = 2 minutes.
- This is the best case in which the vehicle crosses through the centre. Chords shorter than the diameter give more frequent handoffs, so the average is somewhat higher.
- Higher speed or smaller cells increase handoff rate; microcells under fast traffic are often overlaid by umbrella macro-cells to limit handoffs.

Q20. Total user capacity of a hierarchical network

A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Given

- Regional centres = 8
- Base stations per centre = 40
- Users per base station = 250

Concept / Formula used

```
Total users = centres × base stations per centre × users per base station
```

Step-by-step solution

Step 1. Total base stations = 8 × 40 = 320.

Step 2. Total users = 320 × 250 = 80,000.

Answer: 80,000 users

Additional information

- Hierarchy: BTS → BSC → MSC → regional/core, each level aggregating the traffic below it.
- Controllers and switches at the regional level must be dimensioned for the aggregated signalling and traffic, not only for one base station.
- Users are not all active at once, so the number of subscribers that can be supported is higher than the simultaneous capacity, thanks to statistical multiplexing.

Q21. Number of cells needed to cover 400 km²

Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R²). Roughly how many cells will the rollout require?

Given

- Region area = 400 km²
- Cell radius R = 400 m = 0.4 km
- Approximate cell area = R² (as instructed)

Concept / Formula used

```
Cell area ≈ R²
Number of cells = region area / cell area
```

Step-by-step solution

Step 1. Cell area = (0.4 km)² = 0.16 km².

Step 2. Number of cells = 400 / 0.16 = 2,500.

Answer: ≈ 2,500 cells

Additional information

- Exact hexagon area = $(3\sqrt{3}/2)\ R^2 = 2.598\ R^2 = 2.598 \times 0.16 = 0.4157$ km², which would require 400 / 0.4157 ≈ 963 cells. The problem explicitly says to approximate the area as R², giving 2,500.
- The R² approximation underestimates cell area by a factor of about 2.6 and therefore overestimates the number of sites. Use the exact hexagon formula for real design.
- Radius of 400 m is typical of dense urban micro-cells.

Q22. FDMA channels from 6 MHz with 30 kHz channels

An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Given

- Total spectrum = 6 MHz = 6,000 kHz
- Channel width = 30 kHz

Concept / Formula used

```
N_channels = B_total / B_channel
```

Step-by-step solution

Step 1. N = 6,000 kHz / 30 kHz = 200 channels.

Answer: 200 channels

Additional information

- 30 kHz is the channel width of the North American D-AMPS/IS-136 system, which divides each into 3 TDMA slots.
- If guard bands or control channels are reserved, fewer than 200 are available for voice.
- In FDD, where each call needs an uplink and a downlink channel, the 6 MHz would yield 100 duplex channels if the pool is shared between both directions.

Q23. Offered traffic in Erlangs

A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Given

- Subscribers U = 1,200
- Call rate per user = 4 calls/hour
- Average holding time = 2.5 min = 2.5/60 hour

Concept / Formula used

```
Traffic per user A_u = λ × H
Total traffic A = U × A_u
```

Step-by-step solution

Step 1. Traffic per user = 4 calls/h × (2.5/60) h = 0.1667 Erlang (about 167 mE).

Step 2. Total traffic = 1,200 × 0.1667 = 200 Erlangs.

Step 3. Alternative: total calls/h = 1,200 × 4 = 4,800; call minutes = 4,800 × 2.5 = 12,000 min = 200 hours of traffic in one hour.

Answer: 200 Erlangs

Additional information

- One Erlang is one channel continuously occupied for an hour. A load of 200 Erlangs therefore needs at least 200 channels; to keep the blocking at about 2% (Erlang B), roughly 215–220 channels are required.

- Typical busy-hour traffic per user is 25–100 mE for mobile networks; 167 mE is quite heavy.
- Erlang-B formula: P_B = (A^N / N!) / Σ(A^k / k!, k = 0…N).

Q24. Maximum users in a 40 MHz band at 400 kHz each

A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Given

- Total bandwidth = 40 MHz = 40,000 kHz
- Per-user bandwidth = 400 kHz
- Carrier frequency 2.5 GHz (does not affect the calculation)

Concept / Formula used

```
Users = total bandwidth / bandwidth per user
```

Step-by-step solution

Step 1. Users = 40,000 kHz / 400 kHz = 100.

Answer: 100 simultaneous users

Additional information

- The carrier frequency (2.5 GHz) only determines the propagation and the band; the user count depends only on bandwidth.
- Counting capacity this way assumes pure FDMA with no frequency reuse, no guard bands and no overhead.
- With spectral-efficiency improvements (OFDMA, MIMO) a cell can serve many more users in the same 40 MHz by dividing resources in both frequency and time.

Q25. Total user capacity of a network of base stations

A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Given

- Users per base station = 600
- Number of base stations = 1,200

Concept / Formula used

```
Network capacity = base stations × users per base station
```

Step-by-step solution

Step 1. Capacity = 1,200 × 600 = 720,000 simultaneous users.

Answer: 720,000 simultaneous users

Additional information

- This is the theoretical sum of the capacity of all base stations; real capacity is lower because traffic is not evenly distributed.
- Interference management and frequency reuse are what allow each of the 1,200 stations to reuse the same spectrum.

- Capacity grows linearly with the number of sites, which is why densification is the most effective (but most expensive) capacity upgrade.